CSE STUDY HUB

Data Structures & Algorithms

UNIT 4 · Sorting and hashing

What this unit is about

This unit builds understanding of **Sorting and hashing**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Sorting concepts; stable, in-place and comparison sorting

2. Bubble sort, selection sort and insertion sort: working and complexity

3. Merge sort: divide-and-conquer, merging and complexity

4. Quick sort: pivot selection, partitioning and performance

5. Compare sorting algorithms by time, space and stability

6. Hash tables, hash functions and load factor

7. Collision resolution: separate chaining and open addressing (linear/quadratic probing, double hashing)

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 4 NOTES · PART 1

Sorting concepts; stable, in-place and comparison sorting

Sorting arranges records by a key. Compare algorithms using time, auxiliary space, stability and in-place behaviour. Merge sort is O(n log n); quicksort averages O(n log n) but can degrade to $O(n^2)$.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Bubble sort, selection sort and insertion sort: working and complexity

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Merge sort: divide-and-conquer, merging and complexity

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Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Sorting and hashing, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 4 NOTES · PART 2

Quick sort: pivot selection, partitioning and performance

Sorting arranges records by a key. Compare algorithms using time, auxiliary space, stability and in-place behaviour. Merge sort is O(n log n); quicksort averages O(n log n) but can degrade to $O(n^2)$.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Compare sorting algorithms by time, space and stability

Sorting arranges records by a key. Compare algorithms using time, auxiliary space, stability and in-place behaviour. Merge sort is O(n log n); quicksort averages O(n log n) but can degrade to $O(n^2)$.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Sorting and hashing, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 4 NOTES · PART 3

Hash tables, hash functions and load factor

A hash function maps a key to a table index. Collisions are handled by chaining or open addressing. Performance depends on a good hash function, load factor and resizing policy.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Collision resolution: separate chaining and open addressing (linear/quadratic probing, double hashing)

A hash function maps a key to a table index. Collisions are handled by chaining or open addressing. Performance depends on a good hash function, load factor and resizing policy.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Sorting and hashing, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Sorting concepts; stable, in-place and comparison sorting** — know the definition, core working and one example.

• **Bubble sort, selection sort and insertion sort: working and complexity** — know the definition, core working and one example.

• **Merge sort: divide-and-conquer, merging and complexity** — know the definition, core working and one example.

• **Quick sort: pivot selection, partitioning and performance** — know the definition, core working and one example.

• **Compare sorting algorithms by time, space and stability** — know the definition, core working and one example.

• **Hash tables, hash functions and load factor** — know the definition, core working and one example.

• **Collision resolution: separate chaining and open addressing (linear/quadratic probing, double hashing)** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Sorting and hashing. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Sorting concepts; stable, in-place and comparison sorting and explain its importance.

Q2. Explain the working/steps involved in Bubble sort, selection sort and insertion sort: working and complexity.

Q3. Differentiate between two related concepts from Sorting and hashing.

Q4. Give one practical application of Collision resolution: separate chaining and open addressing (linear/quadratic probing, double hashing) and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Sorting and hashing.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.